

Oli Gurmessa

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EDUCATION

University of St. Thomas

B.S. Computer Science (Fast-Track M.S. track)

Current Student

St. Paul, MN

Coursework: Machine Learning*, Algorithms, Data Structures, Linear Algebra, Probability, Multivariable Calculus

TECHNICAL SKILLS

Languages: Python, Java, C++, TypeScript, LaTeX

ML/DS: PyTorch, TensorFlow/Keras, scikit-learn, NumPy, pandas, OpenCV

Tools: Linux, Git, SQL, Docker, GCP, Power BI

RESEARCH EXPERIENCE

Diffusion Model Probing: Quality and Step Choice Optimization

Computer Science Department — UST

May 2025 – Present

Work in Progress

- Building evaluation harness for diffusion quality vs. inference steps using controlled prompts/seeds and PickScore.
- Running **1,600 prompts** × **27 step configs**; refining methodology to quantify quality-compute tradeoffs.
- Exploring compute reduction of **5–10%** while maintaining **PickScore 0.90–0.95**; documenting reproducible protocols.

AI Audio for User-Generated Content in Video Games

Computer Science Department — UST

May 2024 – Sep 2024

[10.48550/arXiv.2404.17018](https://arxiv.org/abs/2404.17018)

- Developed text-to-audio/image-to-audio pipelines for dynamic game audio; refined approach through iterative team feedback.
- Converted **50+ levels** and **30+ configs** into prompts for MusicGen/AudioGen; optimized pipeline to **4s** per clip.
- Learned to establish evaluation metrics (quality, controllability, latency); compared model variants for aesthetic alignment.

GeezNet: Deep Architectures for Ge'ez Numeral Recognition

Undergraduate Research Opportunities Program — UST

May 2023 – Sep 2023

[10.1109/TVCG.2024.3456345](https://arxiv.org/abs/2404.34563)

- Implemented CNN/ResNet50/InceptionV3 baselines for Ge'ez numeral recognition; did preprocessing/augmentation.
- Achieved **97.8% accuracy** and **98% F1** on **10,000-image** dataset; performed error analysis on confusing character pairs.
- Ran ablation studies on augmentation/regularization; documented reproducible configurations for future research.

WORK EXPERIENCE

Student Technician

UST Innovation and Technology Services

Jan 2023 – Present

St. Paul, MN

- Handle **30–35 weekly tickets** for the university community; troubleshoot issues from MFA/identity to OS/software bugs.
- Collaborate with senior technicians to resolve technical issues, learn new troubleshooting methods to reduce escalations.
- Document repeatable steps to solutions to improve ticket resolution speed and consistency.

Externship – Data Analytics & Marketing

Microsoft GroupMe

Jan 2023 – Apr 2023

Remote

- Learned data analysis with Python and applied it to marketing tasks, creating visualizations to guide engagement decisions.
- Analyzed insights from **120+ survey responses** to guide improvements in university engagement.
- Improved data storytelling using mentor feedback; learned to communicate technical insights clearly to stakeholders.

PROJECTS

SoulSpect – Privacy-First Personal Intelligence App

Personal Project

- Audited and refined lines of code by leveraging AI coding agents to accelerate development
- Built and shipped Next.js app using open-weight MaaS APIs and open-source tools to give AI insights for a persons' daily.
- Created multimodal moment entry; used vector embeddings, ChromaDB, RAG for semantic search across text/audio/images.
- Architected secure storage patterns and data controls for scaling the pipeline to an on-device inference.

HallPals – Residence Life Operations App

Personal Project

- Audited and refined lines of code by leveraging AI coding agents to accelerate development
- Developed AI-assisted RA duty platform with role-based access and incident guidance; refined features through pilot testing.
- Tested with **10 RAs** across **one residence hall**; observed **30-min** efficiency gain; iterated based on user feedback.

From-Scratch Drowsiness Detection (NumPy Neural Network)

Course Project

- Implemented neural network **from scratch in NumPy** to classify eye states; achieved **92% accuracy** via iterative tuning.
- Built forward/backward props with matrix ops, ReLU/Softmax, and gradient descent; solidified backprop understanding.

LEADERSHIP

Residence Advisor, UST Residence Life

Jan 2024 – Present

- Mentored **40+ residents** in community building and conflict mediation; completed duty rounds and crisis response.

Vice President, Lead Club

Jan 2023 – Jan 2024

- Led leadership workshops and mentorship programs through executive board collaboration.

Event Coordinator, Computer Science Club

Jan 2023 – Sep 2023

- Organized **8+ technical workshops** and networking events using feedback and attendance data.